CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD

CENTRAL VALLEY REGION
11020 SUN CENTER DR #200
RANCHO CORDOVA CA 95670-6114

2023 ANNUAL REPORT

PREPARED FOR:

FANELLI DAIRY-HEIFERS

23839 W BLOSS AVE

HILMAR CA 95324

NOTES:

PREPARED BY:

CARDOSO AG SERVICES INC.
PO BOX 906
NEWMAN CA 95360

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Fanelli Dairy-Heifers

Physical address of dairy:

23839 W Bloss AVE	Hilmar	Merced	95324
Number and Street	City	County	Zip Code

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1960

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X045-X190-X058-XXXX X045-X190-X070-XXXX

B. OPERATORS

Fanelli, Victor

Operator name: Fanelli, Victor Telephone no.: Landline (209) 678-2858 Cellular

7689 Washington Rd	Hilmar	CA	95324
Mailing Address Number and Street	City	State	Zip Code

This operator is responsible for paying permit fees.

C. OWNERS

Fanelli, Victor

Legal owner name: Fanelli, Victor Telephone no.: Landline (209) 678-2858 Cellular

7689 Washington Rd	Hilmar	CA	95324
Mailing Address Number and Street	City	State	Zip Code

This owner is responsible for paying permit fees.

Fanelli Brothers

Legal owner name:	Fanelli Brothers	Telephone no.:		(209) 678-3137
			Landline	Cellular
7689 Washington Rd		Hilmar	CA	95324
Mailing Address Number and Street		City	State	Zip Code

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	0	0	0	0
Number under roof	0	0	0	153	148	0
Maximum number	0	0	0	153	148	0
Average number	0	0	0	153	148	0
Avg live weight (lbs)	0	0	0	500		

Predominant milk cow breed: Jersey

Average milk production: 1 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 1,861.28 *tons per reporting period*

Total nitrogen from manure: 22,082.50 *lbs per reporting period*

After ammonia losses (30% loss applied): 15,457.75 *lbs per reporting period*

Total phosphorus from manure: 2,991.98 *lbs per reporting period*

Total potassium from manure: 1.00 *lbs per reporting period*

Total salt from manure: 0.00 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 5,508,000 *gallons*

Total nitrogen generated: 11,891.67 *lbs*

Total phosphorus generated: 9,506.80 *lbs*

Total potassium generated: 52,576.50 *lbs*

Total salt generated: 228,432.36 *lbs*

	5,508,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	5,508,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Canal	Surface water
Domestic Well	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

Date	Material type / Description	Quantity	Reporting basis	Moist. (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
01/01/2023	Separator solids *Solid Manure Import*	288.00 *ton*	Dry-weight	74.1	16,400.00	3,000.00	4,300.00		14.30
04/29/2023	Separator solids *Solid Manure Import*	670.00 *ton*	Dry-weight	74.1	16,400.00	3,000.00	4,300.00		14.30

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	8,138.40	1,488.73	2,133.85	70,962.89
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	8,138.40	1,488.73	2,133.85	70,962.89

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
The 10	12	12	3	both	X045-X190-X058-XXXX
The 40	43	43	2	both	X045-X190-X058-XXXX X045-X190-X070-XXXX
Totals for areas that were used for application	55	55	5		
Totals for areas that were not used for application					
Land application area totals	55	55	5		

B. CROPS AND HARVESTS

The 10

Field name: The 10

09/16/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 12 Plant date: 09/16/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/28/2023	177.00 *ton*	Dry-weight		65.8	23,200.00	2,900.00	18,900.00		10.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	14.75	234.06	29.26	190.68	1,049.26

05/17/2023: Corn, silage

Crop: Corn, silage Acres planted: 12 Plant date: 05/17/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/06/2023	270.00 *ton*	Dry-weight		63.4	11,700.00	3,100.00	20,500.00		6.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	22.50	192.70	51.06	337.64	1,087.02

The 10

09/06/2023: Sudangrass, silage

Crop: Sudangrass, silage Acres planted: 12 Plant date: 09/06/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
11/01/2023	88.00 *ton*	Dry-weight		66.4	12,900.00	2,750.00	20,200.00		6.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	88.00	13.60	96.00	0.00
Total actual harvest content	7.33	63.57	13.55	99.55	330.18

The 40

Field name: The 40

09/16/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 43 Plant date: 09/16/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/28/2023	634.00 *ton*	Dry-weight		77.8	14,400.00	2,900.00	39,000.00		11.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	14.74	94.27	18.98	255.31	765.93

05/17/2023: Corn, silage

Crop: Corn, silage Acres planted: 43 Plant date: 05/17/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/06/2023	967.00 *ton*	Dry-weight		69.4	14,100.00	2,400.00	19,900.00		6.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	22.49	194.06	33.03	273.88	922.11

NUTRIENT BUDGET

A. LAND APPLICATIONS

The 10 - 09/16/2022: Oats, silage-soft dough

Field name: The 10

Crop: Oats, silage-soft dough — Plant date: 09/16/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/25/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Imported Solid Manure	Separator solids	203.88	37.30	53.46	1,777.78	288.00 *ton*
Application event totals		203.88	37.30	53.46	1,777.78	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/06/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	38.14	117.25	194.82	846.64	261,818.00 *gal*
Canal	Surface water	7.40	0.00	0.00	0.00	1,662,723.00 *gal*
Application event totals		45.54	117.25	194.82	846.64	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	74.23	12.40	234.87	992.30	261,818.00 *gal*
Application event totals		74.23	12.40	234.87	992.30	

The 10 - 05/17/2023: Corn, silage

Field name: The 10

Crop: Corn, silage — Plant date: 05/17/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/29/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Imported Solid Manure	Separator solids	103.36	18.91	27.10	901.23	146.00 *ton*
Application event totals		103.36	18.91	27.10	901.23	

The 10 - 05/17/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	8.84	0.00	0.00	0.00	1,986,004.00 *gal*
Application event totals		8.84	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	43.97	18.74	228.39	982.90	293,236.00 *gal*
Canal	Surface water	7.78	0.00	0.00	0.00	1,747,409.00 *gal*
Application event totals		51.74	18.74	228.39	982.90	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	13.49	0.00	0.00	0.00	3,030,827.00 *gal*
Application event totals		13.49	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	11.22	0.00	0.00	0.00	2,519,977.00 *gal*
Application event totals		11.22	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	11.78	0.00	0.00	0.00	2,646,232.00 *gal*
Application event totals		11.78	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	9.14	0.00	0.00	0.00	2,053,469.00 *gal*
Application event totals		9.14	0.00	0.00	0.00	

The 10 - 05/17/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	43.97	18.74	228.39	982.90	293,236.00 *gal*
Canal	Surface water	10.83	0.00	0.00	0.00	2,433,214.00 *gal*
Application event totals		54.79	18.74	228.39	982.90	

The 10 - 09/06/2023: Sudangrass, silage

Field name: The 10

Crop: Sudangrass, silage

Plant date: 09/06/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	72.90	26.20	318.36	1,597.65	420,000.00 *gal*
Canal	Surface water	10.58	0.00	0.00	0.00	2,377,180.00 *gal*
Application event totals		83.48	26.20	318.36	1,597.65	

The 40 - 09/16/2022: Oats, silage-soft dough

Field name: The 40

Crop: Oats, silage-soft dough

Plant date: 09/16/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/06/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	38.14	117.25	194.82	846.64	938,182.00 *gal*
Canal	Surface water	7.40	0.00	0.00	0.00	5,958,090.00 *gal*
Application event totals		45.54	117.25	194.82	846.64	

The 40 - 09/16/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	74.23	12.40	234.87	992.30	938,182.00 *gal*
Application event totals		74.23	12.40	234.87	992.30	

The 40 - 05/17/2023: Corn, silage

Field name: The 40

Crop: Corn, silage

Plant date: 05/17/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/29/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Imported Solid Manure	Separator solids	103.52	18.94	27.14	902.67	524.00 *ton*
Application event totals		103.52	18.94	27.14	902.67	

05/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	8.84	0.00	0.00	0.00	7,116,514.00 *gal*
Application event totals		8.84	0.00	0.00	0.00	

06/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	43.97	18.74	228.39	982.90	1,050,764.00 *gal*
Canal	Surface water	7.78	0.00	0.00	0.00	6,261,550.00 *gal*
Application event totals		51.74	18.74	228.39	982.90	

07/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	13.49	0.00	0.00	0.00	10,860,465.00 *gal*
Application event totals		13.49	0.00	0.00	0.00	

The 40 - 05/17/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	11.22	0.00	0.00	0.00	9,029,919.00 *gal*
Application event totals		11.22	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	11.78	0.00	0.00	0.00	9,482,332.00 *gal*
Application event totals		11.78	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	9.14	0.00	0.00	0.00	7,358,263.00 *gal*
Application event totals		9.14	0.00	0.00	0.00	

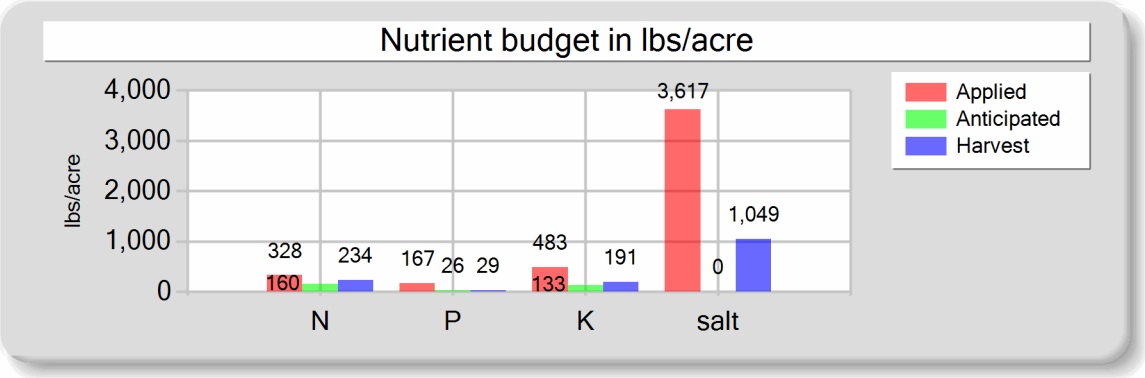
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	43.97	18.74	228.39	982.90	1,050,764.00 *gal*
Canal	Surface water	10.83	0.00	0.00	0.00	8,719,018.00 *gal*
Application event totals		54.79	18.74	228.39	982.90	

B. NUTRIENT BUDGET

The 10 - 09/16/2022: Oats, silage-soft dough

Field name: The 10 Crop: Oats, silage-soft dough Plant date: 09/16/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	203.88	37.30	53.46	1,777.78
Process wastewater	112.38	129.65	429.69	1,838.93
Fresh water	7.40	0.00	0.00	0.00
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	328.33	166.95	483.15	3,616.71
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	234.06	29.26	190.68	1,049.26
Nutrient balance	94.26	137.69	292.47	2,567.45
Applied to removed ratio	1.40	5.71	2.53	3.45

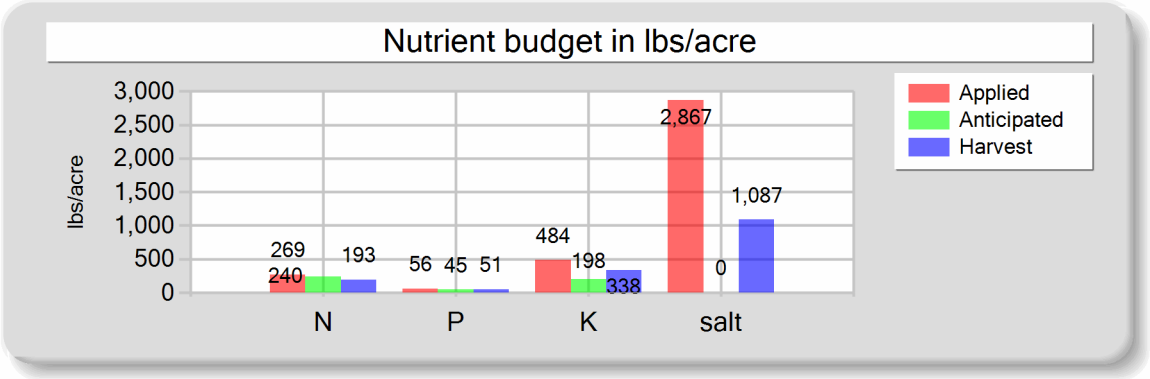
Fresh water applied	
1,662,723.00	*gallons*
61.23	*acre-inches*
5.10	*inches/acre*

Process wastewater applied	
523,636.00	*gallons*
19.28	*acre-inches*
1.61	*inches/acre*

Total harvests for the crop	
1	*harvests*

The 10 - 05/17/2023: Corn, silage

Field name: The 10 Crop: Corn, silage Plant date: 05/17/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	103.36	18.91	27.10	901.23
Process wastewater	87.93	37.48	456.78	1,965.80
Fresh water	73.07	0.00	0.00	0.00
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	269.02	56.39	483.88	2,867.03
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	192.70	51.06	337.64	1,087.02
Nutrient balance	76.32	5.33	146.25	1,780.01
Applied to removed ratio	1.40	1.10	1.43	2.64

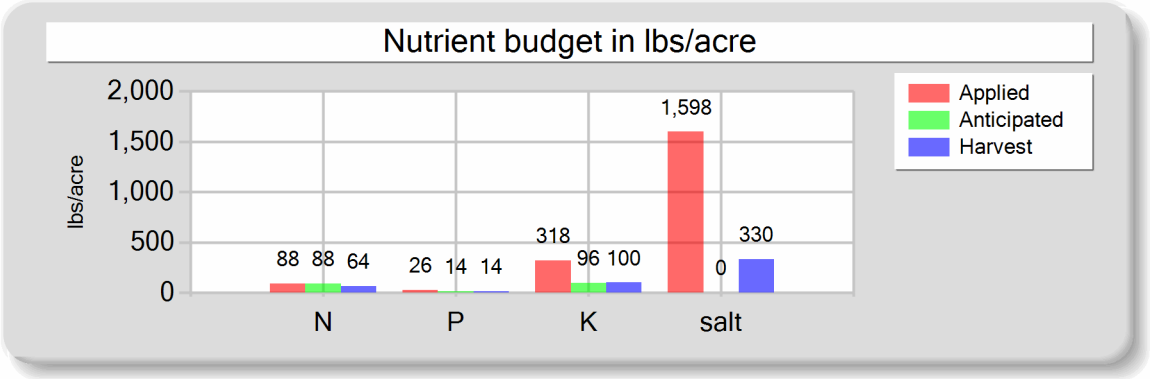
Fresh water applied	
16,417,132.00	*gallons*
604.59	*acre-inches*
50.38	*inches/acre*

Process wastewater applied	
586,472.00	*gallons*
21.60	*acre-inches*
1.80	*inches/acre*

Total harvests for the crop	
1	*harvests*

The 10 - 09/06/2023: Sudangrass, silage

Field name: The 10 | Crop: Sudangrass, silage | Plant date: 09/06/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	72.90	26.20	318.36	1,597.65
Fresh water	10.58	0.00	0.00	0.00
Atmospheric deposition	4.67	0.00	0.00	0.00
Total nutrients applied	88.15	26.20	318.36	1,597.65
Anticipated crop nutrient removal	88.00	13.60	96.00	0.00
Actual crop nutrient removal	63.57	13.55	99.55	330.18
Nutrient balance	24.58	12.65	218.82	1,267.47
Applied to removed ratio	1.39	1.93	3.20	4.84

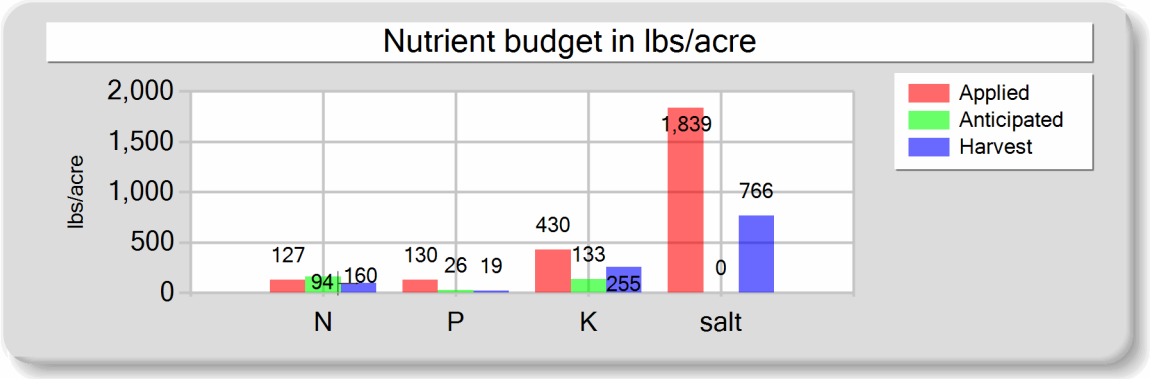
Fresh water applied	
2,377,180.00	*gallons*
87.54	*acre-inches*
7.30	*inches/acre*

Process wastewater applied	
420,000.00	*gallons*
15.47	*acre-inches*
1.29	*inches/acre*

Total harvests for the crop	
1	*harvests*

The 40 - 09/16/2022: Oats, silage-soft dough

Field name: The 40 Crop: Oats, silage-soft dough Plant date: 09/16/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	112.38	129.65	429.69	1,838.93
Fresh water	7.40	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	126.78	129.65	429.69	1,838.93
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	94.27	18.98	255.31	765.93
Nutrient balance	32.51	110.67	174.38	1,073.00
Applied to removed ratio	1.34	6.83	1.68	2.40

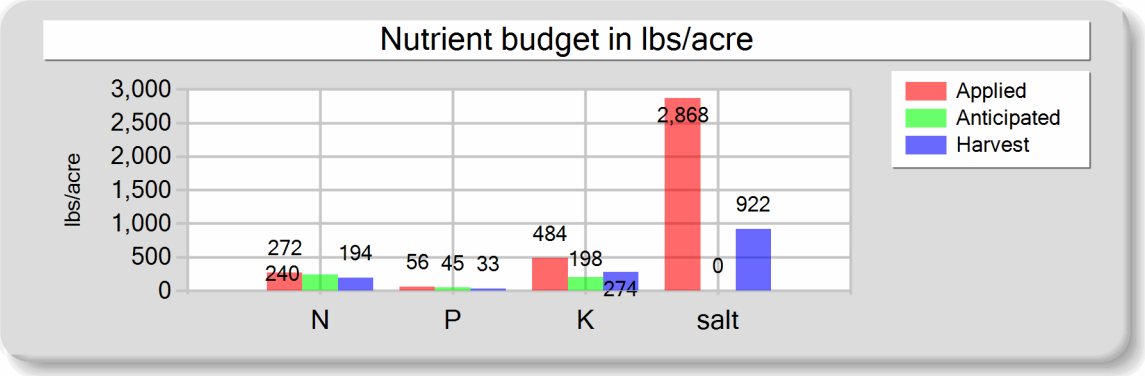
Fresh water applied	
5,958,090.00	*gallons*
219.42	*acre-inches*
5.10	*inches/acre*

Process wastewater applied	
1,876,364.00	*gallons*
69.10	*acre-inches*
1.61	*inches/acre*

Total harvests for the crop	
1	*harvests*

The 40 - 05/17/2023: Corn, silage

Field name: The 40 | Crop: Corn, silage | Plant date: 05/17/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	103.52	18.94	27.14	902.67
Process wastewater	87.93	37.48	456.78	1,965.80
Fresh water	73.07	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	271.52	56.42	483.93	2,868.47
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	194.06	33.03	273.88	922.11
Nutrient balance	77.46	23.39	210.05	1,946.36
Applied to removed ratio	1.40	1.71	1.77	3.11

Fresh water applied	
58,828,061.00	*gallons*
2,166.44	*acre-inches*
50.38	*inches/acre*

Process wastewater applied	
2,101,528.00	*gallons*
77.39	*acre-inches*
1.80	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Corral Manure

Sample and source description: Corral Manure

Sample date: 05/08/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 60.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,600.00	13,400.00	5,300.00	43,100.00	14,100.00	900.00	5,600.00	0.00		68.60
DL	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01		1.00

Corral Manure

Sample and source description: Corral Manure

Sample date: 11/14/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 43.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,700.00	4,500.00	17,800.00							0.00
DL	0.01	0.01	0.01							1.00

B. PROCESS WASTEWATER ANALYSES

2022 Lagoon Q4

Sample and source description: 2022 Lagoon Q4

Sample date: 11/03/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: 8.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	208.00	70.80	0.00	1.50	644.00	1,070.00								6,890.00	4,650
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Lagoon Q1

Sample and source description: Lagoon Q1

Sample date: 03/07/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 8.10

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	406.00	253.00	0.00	1.70	68.10	1,290.00								7,920.00	5,450
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Lagoon Q2

Sample and source description: Lagoon Q2

Sample date: 05/09/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.80

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	396.00	168.00	0.00	1.00	112.00	1,840.00								9,520.00	6,400
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Lagoon Q3

Sample and source description: Lagoon Q3

Sample date: 07/26/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 8.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	215.00	96.70	0.00	0.60	91.90	1,120.00								6,720.00	4,820
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Lagoon Q4

Sample and source description: Lagoon Q4

Sample date: 11/14/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 8.30

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	249.00	81.50	0.00	0.60	89.70	1,090.00								6,760.00	5,470
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

C. FRESH WATER ANALYSES

Canal

TID Canal

Sample description: TID Canal

Sample date: 09/21/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	6.40	0.00	6.40								284.00	0
DL	0.01	0.01	0.01								10.00	1

Domestic Well

DW MB

Sample description: DW MB

Sample date: 12/11/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	46.70	0.00	46.70								2,150.00	0
DL	0.01	0.01	0.01								10.00	1

D. SOIL ANALYSES

The 10

The 10 Soil

Sample and source description: The 10 Soil

Sample date: 09/07/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			28.00				
DL			0.01				

The 40

The 40 Soil

Sample and source description: The 40 Soil

Sample date: 09/07/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			24.00				
DL			0.01				

E. PLANT TISSUE ANALYSES

The 10 - 09/16/2022: Oats, silage-soft dough

Small Field Oats

Sample and source description: Small Field Oats

Sample date: 04/28/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	23,200.00	2,900.00	18,900.00		10.40
DL	0.01	0.01	0.01		1.00

The 10 - 05/17/2023: Corn, silage

Small Field Corn

Sample and source description: Small Field Corn

Sample date: 09/06/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 63.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,700.00	3,100.00	20,500.00		6.60
DL	0.01	0.01	0.01		1.00

The 10 - 09/06/2023: Sudangrass, silage

Sudan Small Field

Sample and source description: Sudan Small Field

Sample date: 11/01/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	21,000.00	3,300.00	31,900.00		11.10
DL	0.01	0.01	0.01		1.00

The 40 - 09/16/2022: Oats, silage-soft dough

Big Field Oats

Sample and source description: Big Field Oats

Sample date: 04/28/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 77.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,400.00	2,900.00	39,000.00		11.70
DL	0.01	0.01	0.01		1.00

The 40 - 05/17/2023: Corn, silage

Big Field Corn

Sample and source description: Big Field Corn

Sample date: 09/06/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,100.00	2,400.00	19,900.00		6.70
DL	0.01	0.01	0.01		1.00

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

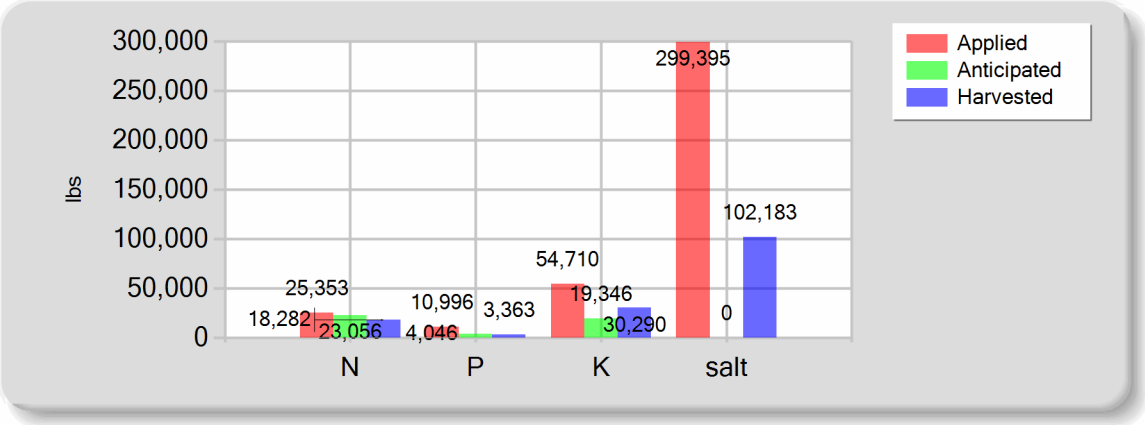
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

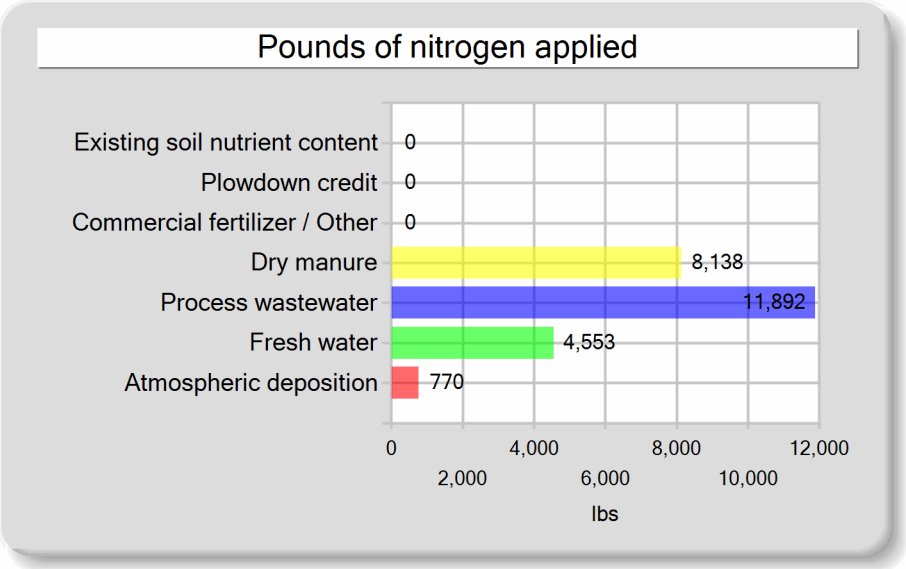
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

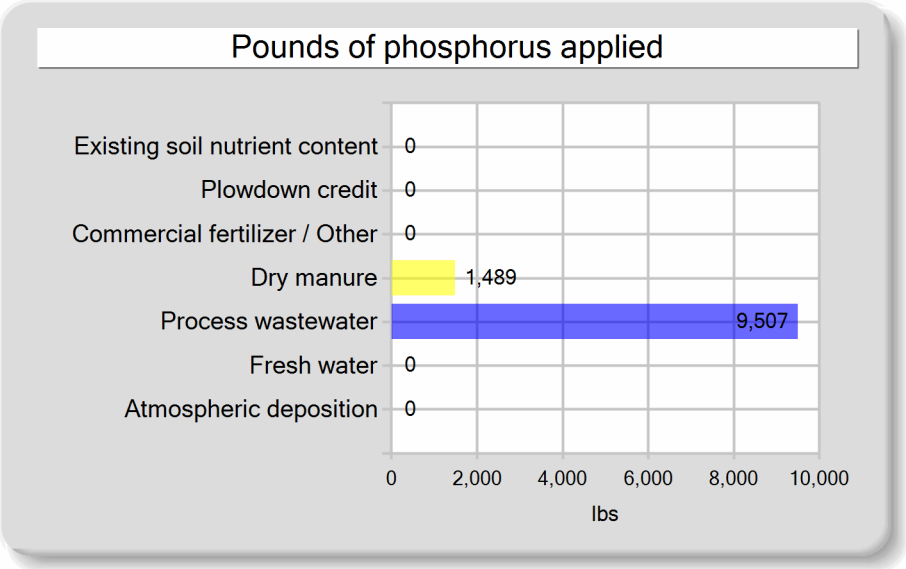
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	8,138.40	1,488.73	2,133.85	70,962.89
Process wastewater	11,891.67	9,506.80	52,576.50	228,432.36
Fresh water	4,552.67	0.00	0.00	0.00
Atmospheric deposition	770.00	0.00	0.00	0.00
Total nutrients applied	25,352.73	10,995.53	54,710.35	299,395.25
Anticipated crop nutrient removal	23,056.00	4,046.20	19,346.00	0.00
Actual crop nutrient removal	18,282.00	3,363.07	30,289.60	102,183.32
Nutrient balance	7,070.74	7,632.46	24,420.76	197,211.93
Applied to removed ratio	1.39	3.27	1.81	2.93

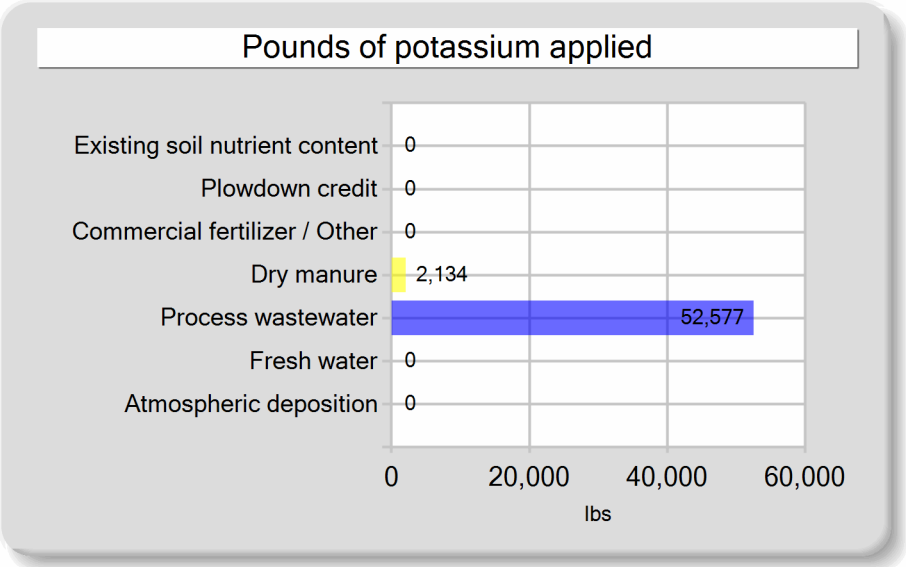
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

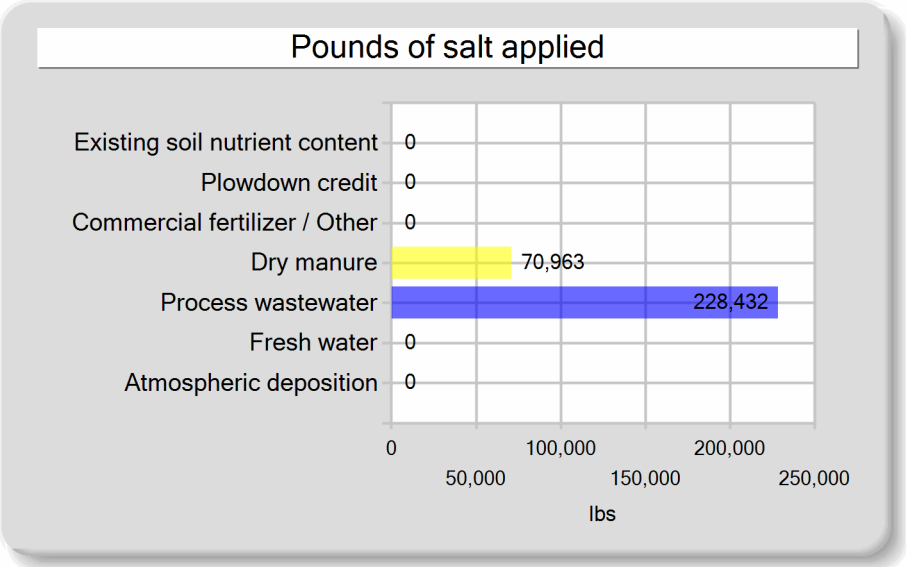


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Victor Fanelli

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Victor Fanelli	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
Jul 1, 2024	
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

DELLAVALLE LABORATORY INC

Report of Dairy Well Water Analysis

Fanelli Dairy
7689 Washington Rd
Hilmar CA 95324
00-0023474 08

E-mail:
Copy To: ;cas.labs@yahoo.com

Lab No.: 23L0735
Sampled By: M Dekeyrel
Requested By: Victor Fanelli
Submitted Date: 12/12/23
Reported Date: 12/21/23
Project: Fanelli Brothers
Crop ID:

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Field NH_4-N mg/L	Total NH_4-N mg/L	pH at 25°C unit
1	DW - MB	12/11/23	11:00	2150	2.15	46.7	ND		7.4

***** = Field NH_4-N not Taken.
ND = None Detected

Approved By: Scott M Friedland
Laboratory Director\Technical Manager
ELAP Certification #1595
A2LA Certification #6440.02

DELLAVALLE LABORATORY INC

Report of Dairy Well Water Analysis

Fanelli Dairy
7689 Washington Rd
Hilmar CA 95324
00-0023474 08

E-mail:
Copy To: ;cas.labs@yahoo.com

Lab No.: 23L0735
Sampled By: M Dekeyrel
Requested By: Victor Fanelli
Submitted Date: 12/12/23
Reported Date: 12/21/23
Project: Fanelli Brothers
Crop ID:

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Nitrogen agricultural use Calcs calculated from nitrate-N lbs/AcFoot	lbs/AcInch
1	DW - MB	12/11/23	11:00	**2150**	**2.15**	**46.7**	127.49	10.62

Water for Crop Use **Nutrient Management**	Total Salts	Total Salts	Nitrate Nitrogen
Low Levels	< 900	< 0.90	< 2.0
Significant Levels	900-2200	0.90-2.2	2.0-10.0
High Levels	2200+	2.2+	10.0+

Sampling abbreviations: H.B. = hose bib, S.P. = stand pipe, Dom. = domestic well, IR = irrigation well, AG = ag well.
Nitrate exceeds Water Quality MCL levels if results are above 10 mg/l nitrate-nitrogen NO3-N (equivalent to 45 mg/l nitrate, NO3).
Total Salt results are used to monitor changes of salt in the well aquifers between annual tests.
MCL = Maximum Contaminant Level according to the California Domestic Water Quality and Monitoring Regulations (Title 22)
MDL = Method Detection Limit; **RL** = Reporting Limit, mg/L = ppm.
SM = Standard Methods for the Examination of Water and Wastewater
EPA = Environmental Protection Agency methods used unless otherwise indicated.

DELLAVALLE™
LABORATORY INC

Fanelli Dairy
7689 Washington Rd
Hilmar, CA 95324

Account# 00-0023474
Account Manager: Ben Nydam
Submitted By: Victor Fanelli
Ranch: Fanelli Brothers

Received: 12/12/2023 15:54
Reported: 12/21/2023 08:40

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23L0735-01	DW - MB	Well Water	M Dekeyrel		12/11/2023 11:00

Default Cooler Temperature on Receipt °C: -0.4
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™
LABORATORY INC

Fanelli Dairy
7689 Washington Rd
Hilmar, CA 95324

Account# 00-0023474
Account Manager: Ben Nydam
Submitted By: Victor Fanelli
Ranch: Fanelli Brothers

Received: 12/12/2023 15:54
Reported: 12/21/2023 08:40

Sample Results

Sample: DW - MB
23L0735-01 (Water)

Sampled: 12/11/2023 11:00
Sampled By: M Dekeyrel

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**2.15**	mmhos/cm	0.01	1		12/13/23 19:02	SM 2510 B		BEL0587
Electrical Conductivity umhos	**2150**	umhos/cm	10.0	1		12/13/23 19:02	SM 2510 B		BEL0587
Ammonia (as N)	ND	mg/L	0.00	1		12/11/23 11:00	Field		BEL0406
Nitrate Nitrogen as NO3N	**46.7**	mg/L	0.1	1	10	12/13/23 09:50	EPA 300.0		BEL0350
Temperature	**25.0**	units	0.0	1		12/13/23 19:02	SM 4500-H+	H	BEL0587
pH	**7.4**	units	1.0	1		12/13/23 19:02	SM 4500-H+	H	BEL0587

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™ LABORATORY INC

Fanelli Dairy
7689 Washington Rd
Hilmar, CA 95324

Account# 00-0023474
Account Manager: Ben Nydam
Submitted By: Victor Fanelli
Ranch: Fanelli Brothers

Received: 12/12/2023 15:54
Reported: 12/21/2023 08:40

Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0350										
Blank (BEL0350-BLK1)					Prepared & Analyzed: 12/12/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0350-BLK2)					Prepared & Analyzed: 12/12/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0350-BLK3)					Prepared: 12/12/2023 Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0350-BLK4)					Prepared: 12/12/2023 Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0350-BLK5)					Prepared: 12/12/2023 Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEL0350-BS1)					Prepared: 12/12/2023 Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	4.6		0.1	mg/L	5.000		92.8	90-110		
LCS (BEL0350-BS2)					Prepared: 12/12/2023 Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	5.4		0.1	mg/L	5.000		107	90-110		
LCS (BEL0350-BS3)					Prepared: 12/12/2023 Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	4.7		0.1	mg/L	5.000		94.3	90-110		
LCS (BEL0350-BS4)					Prepared: 12/12/2023 Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	4.6		0.1	mg/L	5.000		92.7	90-110		
Duplicate (BEL0350-DUP1)		**Source: 23L0592-01**			Prepared: 12/12/2023 Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	0.08		0.1	mg/L		0.08			0.00	10
Duplicate (BEL0350-DUP2)		**Source: 23L0594-01**			Prepared: 12/12/2023 Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	0.09		0.1	mg/L		0.09			5.65	10
Duplicate (BEL0350-DUP3)		**Source: 23L0724-01**			Prepared: 12/12/2023 Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	0.6		0.1	mg/L		0.6			1.23	10
Duplicate (BEL0350-DUP4)		**Source: 23L0731-01**			Prepared: 12/12/2023 Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	0.2		0.1	mg/L		0.2			0.466	10
Matrix Spike (BEL0350-MS1)		**Source: 23L0592-01**			Prepared & Analyzed: 12/12/2023					
Nitrate Nitrogen as NO3N	4.6		0.1	mg/L	5.000	0.08	90.7	90-110		
Matrix Spike (BEL0350-MS2)		**Source: 23L0594-01**			Prepared: 12/12/2023 Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	4.7		0.1	mg/L	5.000	0.09	91.7	90-110		

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™ LABORATORY INC

Fanelli Dairy
7689 Washington Rd
Hilmar, CA 95324

Account# 00-0023474
Account Manager: Ben Nydam
Submitted By: Victor Fanelli
Ranch: Fanelli Brothers

Received: 12/12/2023 15:54
Reported: 12/21/2023 08:40

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0350 (Continued)										
Matrix Spike (BEL0350-MS4)			**Source: 23L0731-01**		Prepared: 12/12/2023 Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	4.9		0.1	mg/L	5.000	0.2	93.7	90-110		
Reference (BEL0350-SRM1)					Prepared & Analyzed: 12/12/2023					
Nitrate Nitrogen as NO3N	9.2			mg/L	10.00		92.0	90-110		
Reference (BEL0350-SRM2)					Prepared: 12/12/2023 Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	9.3			mg/L	10.00		92.6	90-110		
Reference (BEL0350-SRM3)					Prepared: 12/12/2023 Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	9.3			mg/L	10.00		92.7	90-110		
Reference (BEL0350-SRM4)					Prepared: 12/12/2023 Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	9.4			mg/L	10.00		93.6	90-110		
Reference (BEL0350-SRM5)					Prepared: 12/12/2023 Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	9.2			mg/L	10.00		92.2	90-110		

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™
LABORATORY INC

Fanelli Dairy
7689 Washington Rd
Hilmar, CA 95324

Account# 00-0023474
Account Manager: Ben Nydam
Submitted By: Victor Fanelli
Ranch: Fanelli Brothers

Received: 12/12/2023 15:54
Reported: 12/21/2023 08:40

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0587										
Blank (BEL0587-BLK1)					Prepared & Analyzed: 12/13/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
pH	5.5		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Blank (BEL0587-BLK2)					Prepared & Analyzed: 12/13/2023					
Temperature	25.0		0.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
pH	7.3		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Blank (BEL0587-BLK3)					Prepared & Analyzed: 12/13/2023					
Temperature	25.0		0.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
pH	7.7		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Duplicate (BEL0587-DUP1)		**Source: 23L0731-02**			Prepared & Analyzed: 12/13/2023					
Electrical Conductivity	0.34		0.01	mmhos/cm		0.33			0.509	10
pH	7.2		1.0	units		7.3			1.66	10
Electrical Conductivity umhos	335		10.0	umhos/cm		333			0.509	10
Duplicate (BEL0587-DUP2)		**Source: 23L0737-03**			Prepared & Analyzed: 12/13/2023					
Electrical Conductivity	0.68		0.01	mmhos/cm		0.66			3.31	10
Electrical Conductivity umhos	682		10.0	umhos/cm		659			3.31	10
pH	8.3		1.0	units		8.3			0.00	10
Reference (BEL0587-SRM1)					Prepared & Analyzed: 12/13/2023					
Electrical Conductivity	448			umhos/cm	426.0		105	90-110		
Reference (BEL0587-SRM2)					Prepared & Analyzed: 12/13/2023					
pH	7.5			units	7.520		100	67021-101.32		
Reference (BEL0587-SRM3)					Prepared & Analyzed: 12/13/2023					
Electrical Conductivity	1080			umhos/cm	1000		108	90-110		
Electrical Conductivity umhos	1080			umhos/cm	1000		108	90-110		
Reference (BEL0587-SRM4)					Prepared & Analyzed: 12/13/2023					
Electrical Conductivity	1070			umhos/cm	1000		107	90-110		
Electrical Conductivity umhos	1070			umhos/cm	1000		107	90-110		
Reference (BEL0587-SRM5)					Prepared & Analyzed: 12/13/2023					
Electrical Conductivity	1060			umhos/cm	1000		106	90-110		

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

DELLAVALLE™
LABORATORY INC

Fanelli Dairy
7689 Washington Rd
Hilmar, CA 95324

Account# 00-0023474
Account Manager: Ben Nydam
Submitted By: Victor Fanelli
Ranch: Fanelli Brothers

Received: 12/12/2023 15:54
Reported: 12/21/2023 08:40

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0587 (Continued)										
Reference (BEL0587-SRM5)					Prepared & Analyzed: 12/13/2023					
Electrical Conductivity umhos	1060			umhos/cm	1000		106	90-110		
Reference (BEL0587-SRM6)					Prepared & Analyzed: 12/13/2023					
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEL0587-SRM7)					Prepared & Analyzed: 12/13/2023					
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEL0587-SRM8)					Prepared & Analyzed: 12/13/2023					
pH	4.0			units	4.000		100	97.5-102.5		

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

12/12/23 15:54 23L0735

JG

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 • 800 228-9896 • Fax 559 268-8174

WATER WORK REQUEST

Bill To: Acct No. 23474 Cons. 8

Purchase Order No. _____ Results Needed By _____

Client: **Fanelli Dairy**
Address: 7689 Washington Road
City, State, Zip: Hilmar CA 95324
Phone _____ Fax _____
Cell/Email _____

Copy to: Cardoso Ag Sevices - cas.labs@yahoo.com

Requested by: Victor Fanelli - (209) 668-8097

Facility: Fanelli Brothers

Date sampled: 12/11/23

Sampled by: M Dekeuvel

[X] QA/QC Document [X] Copy of Chain [ ] RWQCB

No. of Samples _____ No. Bottles _____

Water Type: [ ] Drinking [ ] Wastewater
[X] Ag Water [ ] Ground Water [ ] Mon. Well
[ ] Supply Water [ ] Other _____

Analysis and Bottles Required: *(Please Indicate Analysis)*

(8) DWW1: (EC, pH, NO_3-N, **NH_4-N Field Test)**
(1) l L plastic, unpreserved (white)

( ) DWW2: (DWW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na, TDS)
(1) l L plastic, unpreserved (white)

( ) DCW1: (EC, NO_3-N, TDS)
(1) l L plastic, unpreserved (white)

( ) DPW1: (EC, pH, NO_3-N, NH_4-N, TKN, TDS, TP, TK)
(1) l L plastic, unpreserved (white)

( ) DPW2: (DPW1 Plus Ca, Mg, Na, HCO_3, CO_3, SO_4, Cl)
(1) l L plastic, unpreserved (white)

( ) Other _____

DESCRIPTION OF SAMPLES		Date Sampled	Time Sampled	Field NH4-N (mg/L)	Received Temp °C
1. DW-MB	Sampled From:	12/11/23	11AM	Ø	-0.4
2.	Sampled From:				
3.	Sampled From:				
4.	Sampled From:				
5.	Sampled From:				
6.	Sampled From:				
7.	Sampled From:				
8.	Sampled From:				
9.	Sampled From:				
10.	Sampled From:				

IR Thermometer SN: 192603727
Correction Factor: 0°C
Calibration Due: 03/06/2024
Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	M Dekeuvel	CASINC	12/11/23 11AM	12/12/23@9:30am
Second	JLP	DLV	12-12-23 9:30	12-12-23 2:01
Third	MM	DLI	12/12 15:54	
Fourth				

I guarantee that as the client, or on behalf of the client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a dated damage fee of 2% per month (annually 24 %) or $5.00 per month whichever is greater.

If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (cal). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through cal under its Rules and Procedures. The prties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration, reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information:

Sampling Hrs ____ Miles ____ Consulting ____

Amt Paid ____ Rec By ____ Check No. ____

Shipping
$ _____ In
$ _____ Out
Date _____

Signature _____

Sample received in cooler with ice?
[ ] Yes [ ] No

ctt:update 2020

Shipping Information: Shipped In ☐ Picked-Up ☒ Walk In ☐ DLI Sampler ☐ Other ☐ _____

Container: Ice Chest ☒ Box ☐ None ☐ **Refrigerant:** Wet Ice ☒ Blue Ice ☐ None ☐

Samples Preserved with HNO_3 or H_2SO_4 were: ☐ Received Preserved ☐ Preserved Upon Receipt at Laboratory

Type of Container(s) Received		Sample Number 1	2	3	4	5	6	7	8	9	10
Sample Containers for Internal (DLI) Use *(Containers that go into the Lab)*											
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic	1									
	250 mL HNO_3 (Red) Plastic										
	* pH Value										
	250 mL H_2SO_4 (Yellow) Plastic										
	* pH Value										
	500 mL unpreserved (White) Plastic										
	1 L unpreserved (White) Plastic										
	1 L unpreserved (BOD) (Purple) Plastic										
"	500mL unpreserved (White) Glass										
	PO4-P Kit										
	Other:										
Sample Containers for Subcontracted ("Send Out") Analyses *(Containers that go in the Subcontract ("Send Out") Refrigerator)*											
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic										
	250 mL HNO_3 (Red) Plastic										
	250 mL H_2SO_4 (Yellow) Plastic										
	500 mL HNO_3 (Red)										
	1 L unpreserved (White) Plastic										
	1 L unpreserved (BOD) (Purple) Plastic										
	1 L HNO_3 (Red)										
VOA Vials	40 mL AG VOA, $Na_2S_2O_3$ + MCAA										
	40 mL AG VOA, $Na_2S_2O_3$										
	40mL AG VOA unpreserved (White) (Set of 3										
	40 mL AG VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
	40mL VOA, H_3PO_4 (Set of 3)										
	40 mL VOA, HCl (Blue) (Set of 3)										
	40 mL VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
Glass	250 mL AG unpreserved (White)										
	250 mL AG H_2SO_4 (Yellow)										
	250 mL AG $Na_2S_2O_3$ (Green)										
	250 mL AG $Na_2S_2O_3$ + MCAA										
	500 mL glass unpreserved (White)										
	500 mL AG HCl (Blue)										
	1 L AG unpreserved (White)										
	1 L AG H_2SO_4 (Yellow)										
	1 L AG $Na_2S_2O_3$ (Green)										
	1 L AG HCl (Blue)										
Special	Cr^{6+} - 50mL Plastic w/Borate/HCO_3/CO_3										
	Cyanide - 500 mL NaOH										
	Asbestos - 1L P wrapped in foil (Set of 2)										
	Sulfide - 1 L **AG** or **P** NaOH + ZnAc										
	Chlorite/Bromate - 250 mL AG with EDA										
	HAA5 - 250mL AG Ammonium Chlorite										
	DO KIT										
	Other:										
	Other:										

12/12/23 15:54 23L0735

SEND OUT